Time complexity of MR prime detector

$n-1 = d \cdot 2^s$ with d odd

$1 \le d \le n-1$

$\therefore 0 \le \log_2 d \le \log_2 (n-1) < \log_2 n \le b$

$\therefore O(\log_2^2 d) = O(b)^2$ — (1) $\qquad \therefore \lceil \log_2 n \rceil = b$

cost of modular multiplication: $M(b) = O(b^2)$

$M(b) = O(b^{1.58})$

FFT $= M(b) = O(b \log b \log \log b)$

Modular exponentiation does $O(\log d)$ modular multiplications

$\therefore$ cost of modular exponentiation : $O(M(b) \cdot \log d)$

pow(a, d, n) $\qquad \approx O(M(b) \cdot b)$ from (1)

TC of check is $O(bM(b) + s\,M(b))$

$\approx O(b\,M(b))$ as $s < b$

TC of is_prime M-R prime test

$= k * b * M(b)$

$= k * b \times b^2$ (naive multiplication)

$= k * b^3 = \boxed{k (\log n)^3}$

Time complexity of my prime computer

$= B * k * (b)^3$

$= n(\log n + \log\log n) * k * b^3$

$\approx n (\log n) * k * (\log n)^3$

$\boxed{\approx n (\log n)^4}$ (k is a constant

Time complexity of Wilsons

$TC = O(j^2)$ as worst case iterations to compute $j!$ is j

TC of orignal ~~wilsons~~ willson's $= \sum_{m=1}^{B} T(\text{wilsons})$

$= \sum_{m=1}^{B} \Theta(m)$

$= \Theta(B^2)$

here $B = 2^n$ (Bertrand's postulate)

worst case $TC = \Theta(4^n)$